Solution

This lens is called convexo-concave lens

Given, $n = 1.52$, $R_1 = 10$ cm and $R_2 = 20$ cm

Both R_1 and R_2 are positive

Lens makers formula, $\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

Substituting the values,

$$\frac{1}{f} = (1.52-1)\left(\frac{1}{10} - \frac{1}{20}\right)$$

$$\frac{1}{f} = (0.52)\left(\frac{2-1}{20}\right) = (0.52)\left(\frac{1}{20}\right) = \frac{0.52}{20}$$

$$f = \frac{20}{0.52} = 38.46\,\text{cm}$$

As the focal length is positive, the lens is a converging lens.

6.6.5 Power of a lens

The power of a lens is a measure of its deviating ability on an incident light. When a ray is incident on a lens then the degree with which the lens deviates the ray is determined by the power of the lens. Power of the lens is inversely proportional to the focal length i.e. greater the power of lens, smaller will be the focal length. In Figure 6.36, the lens (b) has greater deviating ability than lens (a). As (b) has greater deviating ability, its focal length is less and vice versa.

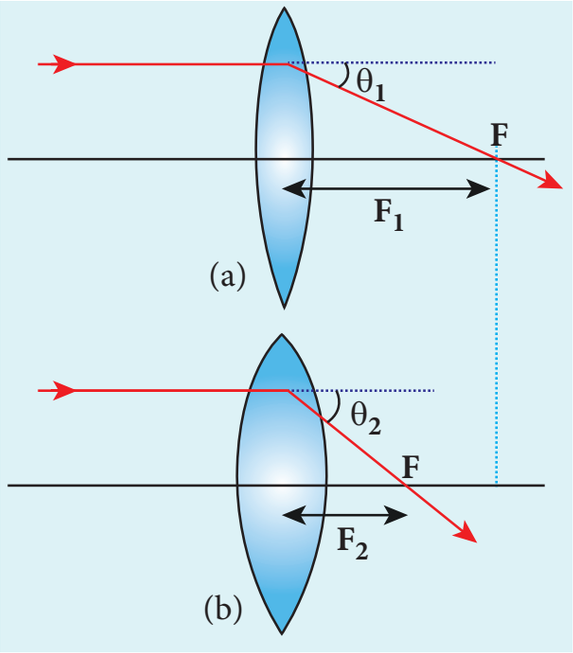


Figure 6.36 Power of lens

In other words, **the *power* of a lens is a measure of the degree of convergence (or) divergence the lens produces on the light falling on it**. The power of a lens P is the reciprocal of its focal length in meter.

$$P = \frac{1}{f} \qquad (6.68)$$

The unit of power is dioptre (D). $1\,\text{D} = 1\,\text{m}^{-1}$. Power is positive for converging lens and negative for diverging lens.

From the lens maker's formula, equation (6.62), the equation (6.68) can be written for power as ,

$$P = \frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad (6.69)$$

The outcome of this equation for power is that for a given geomentry of the lens, larger the value of refractive index, greater is the power of lens and vice versa. Also for lenses with small radii of curvature (bulky) the power is large and for lenses with large radii of curvature (skinny), the power is small.

EXAMPLE 6.15

If the focal length is 150 cm for a lens, what is the power of the lens?

Solution

Given, focal length, $f = 150$ cm $= 1.5$ m

Equation for power of lens is, $P = \frac{1}{f}$

Substituting the values,

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